

Customer Support Ticket Analyzer Using Python

Problem Statement:

Customer support departments receive a large number of service tickets every day from customers reporting technical issues, complaints, feedback, and requests for assistance. Manually reviewing these tickets is time-consuming and inefficient. Organizations need an automated system to organize, clean, analyse, and extract meaningful insights from customer support tickets.

The Customer Support Ticket Analyser is a Python-based project designed to manage support ticket data, clean issue descriptions, analyse customer feedback, identify frequent problems, and generate useful business insights. The system also helps in understanding ticket priorities and customer satisfaction levels through text analysis.

Objectives:

- To store and manage customer support ticket information efficiently.
- To perform text cleaning and preprocessing on issue descriptions.

- To analyse customer feedback using keyword-based analysis.
- To identify the number of High, Medium, and Low priority tickets.
- To detect commonly used words and service-related issues.
- To find the ticket with the longest issue description.
- To extract unique words from customer complaints and feedback.
- To generate meaningful insights that help improve customer support quality.

Types of Analysis Performed:

1. Ticket Data Management

- Stores ticket number, customer name, issue description, and priority.
- Displays all ticket details clearly.

2. Dynamic Ticket Addition

- Allows users to add new support tickets.
- Automatically generates ticket numbers.
- Validates priority input (High/Medium/Low).

3. Text Cleaning Analysis

The issue descriptions are cleaned using:

- Lowercase conversion
- Removal of punctuation symbols
- Removal of extra spaces

- Standardization of words (example: "ok" → "okay")

This improves accuracy during text analysis.

4. Keyword-Based Analysis

The program checks how many tickets contain important keywords such as:

- poor
- good
- slow
- excellent

This helps identify:

- Customer satisfaction
- Service quality
- Common complaints

5. Priority Analysis

Counts the number of:

- High priority tickets
- Medium priority tickets
- Low priority tickets

This helps support teams focus on urgent issues first.

6. Longest Issue Description Analysis

Finds:

- The ticket with the longest complaint description

- Customer details
- Word count

7. Unique Word Analysis

Extracts all unique words from issue descriptions and sorts them alphabetically.

This helps:

- Identify common issue patterns
- Understand customer language usage
- Improve future ticket categorization

Key Insights:

1. Multiple customers reported issues related to:
 - slow service
 - poor handling
 - internet problems
2. Positive feedback words like:
 - good
 - excellent
 - helpfulindicate customer satisfaction in several tickets.
3. High priority tickets represent urgent customer concerns and require immediate action.
4. Text cleaning improves analysis accuracy by removing unnecessary symbols and formatting inconsistencies.

5. Keyword analysis helps management quickly understand overall service quality without manually reading every ticket.
6. Unique word extraction provides better understanding of recurring customer concerns and commonly used terms.
7. Long issue descriptions may indicate complicated technical problems that need deeper investigation.

Conclusion:

The Customer Support Ticket Analyser demonstrates how Python can be used to analyse customer service tickets efficiently. The project manages tickets, cleans customer feedback, performs keyword and priority analysis, and extracts useful insights.

It helps organizations identify common issues, prioritize urgent tickets, improve customer support quality, and make better service decisions using data analysis techniques.